

Title: Leveraging Supervised Machine Learning Techniques in Detecting Anomalies in Food Price Commodity: Insights for Kenya's Market Stability

Dissertation Revision based on Internal and External Examiners' Comments/Suggestions for Revision – Student ID: 190586

Issue Raised	Revision / Improvement Made	Page/para
1. You mentioned supervised learning for anomalies. So that assumes that the anomalies are labelled. So what were the labelling procedures for the anomalies, and how did you prevent that data leakage? What was the criterion for a price being an anomaly? How do you ensure genuine market anomalies other than predicting where a category would fall in the threshold you have defined?	- Added a dedicated Section 3.2.2 Anomaly Labelling Criteria explaining statistical thresholding, contextual validation, and binary class generation. - Clarified that labels were generated using Z-score and IQR thresholds, rolling temporal analysis, and validation against historical market disruptions and food security conditions.	Pg. 26, sec. 3.2.2
2. The first research question specifically mentions inter-commodity relationships, but the results discussion does not clearly show how such relationships were modelled other. There is mention of county-level correlation, multiple commodities, and markets, but limited evidence that inter-commodity dependencies were explicitly engineered. For example, the thesis could have included cross-commodity lag features, commodity-level correlation features, or shared market shock indicators.	- Added discussion explaining that climatic effects are indirect and temporally lagged, particularly within a contemporaneous logistic regression framework. - Expanded discussion to show how supply availability, transport costs, and market dynamics mediate climate impacts on prices. - Clarified that interdependencies were partially captured through engineered features such as lagged prices, rolling volatility, and FPMA reference prices.	Pg. 66, sect.6.6. para. 2, 3& 4
3. The thesis finds that climatic variables such as rainfall, NDVI anomaly, and SPI have lower direct importance on pricing but the student does not	- Added a dedicated discussion explaining that climatic variables influence prices indirectly through production and supply pathways rather than immediate contemporaneous price effects. - Clarified that the logistic regression model estimates direct contemporaneous effects, while climatic shocks	Pg. 66; section 6.6; para:2, 3&4; pg.

provide a rebuttal as to why and what other data would surface this explanation for example market dynamics that capture supply and availability	operate with structural time lags through harvest cycles and supply availability. - Expanded interpretation to show that climate impacts are mediated by market-level dynamics such as supply inflows, storage, transport costs, and distribution constraints. - proposed inclusion of supply-side and market-functioning variables such as market arrival volumes, stock levels, inter-market price spreads, import/export flows, and logistics indicators.	75; section 7.5 para.4
4. The thesis should be careful with the claim that supervised machine learning captures “complex and nonlinear patterns,” Page 68 because the final selected model is Logistic Regression. Logistic Regression can perform well with strong feature engineering, but it is not itself a nonlinear learner unless nonlinearities are introduced through engineered features, interactions, transformations, or categorical encodings.	- Revised wording to clarify that logistic regression is a linear estimator and that complexity arises from engineered temporal and transformed features rather than inherent model nonlinearity.	Pg.72, sec.7.2 para:2
5. Insufficient justification for stratified cross-validation	Added methodological justification for preserving the anomaly class distribution and reducing sampling bias.	Pg. 33; section 3.3, para. 1
6. Weak justification for ROC-optimized thresholds	Explained why ROC-based threshold optimisation was preferred over the default 0.5 threshold in imbalanced anomaly detection settings.	Pg. 33; section 3.3, para. 2
7. Limited explanation of hyperparameter selection	Added discussion showing how model-specific hyperparameters were tuned to balance generalisation and overfitting control.	Pg. 31; section 3.2.10, para.1&2
8. Literature review lacked critical evaluation (dummy, scalability, assumptions)	- Incorporated evidence-based critiques discussing limitations of seasonal decomposition methods, correlational climate studies, and statistical anomaly frameworks.	Pg.5; Subsection 2.2.1, para. 2
9. Lack of critique on seasonal modelling approaches	- Added criticism that dummy-based seasonal methods underestimate asymmetric price fluctuations in African staple markets.	Pg.5; Subsection 2.2.1, para. 1
10. Weak critique of climate-price literature	- Added evidence that many studies rely on correlational relationships and aggregated spatial	Pg.5; Subsection

	data, limiting causal interpretation and local market sensitivity.	2.2.1, para. 3
11. Limited critique of gradient-boosting studies	- Added discussion that gradient-boosted models often lose predictive reliability over longer horizons and in unstable crisis environments.	Page 9; section 2.2.3, para. 5
12. Limited critique of IFPA anomaly frameworks	- Added discussion that purely statistical anomaly thresholds may overlook structural market drivers such as trade policy and speculative dynamics.	Page 9; section 2.2.3, para. 5
13. Abstract improvement Multiple models, including Logistic Regression, Support Vector Machines, Decision Trees, XGBoost and LightGBM were trained and evaluated using stratified cross-validation. The results demonstrated that ensemble models achieve near-perfect recall in detecting anomalies, while Logistic Regression provided the best balance between precision, recall and interpretability, making it the most suitable for deployment.	- Multiple models, including Logistic Regression, Support Vector Machines (SVM), Decision Trees, XGBoost and LightGBM, were trained and evaluated using stratified cross-validation, hyperparameter optimisation and ROC-based threshold calibration to address class imbalance and improve anomaly discrimination. Model performance was assessed using precision, recall, F1-score, true positive rate, false positive rate, and area under the ROC curve (AUC). The results showed that ensemble models achieved strong anomaly detection capability, particularly in recall performance, while Logistic Regression provided the best balance between precision, recall, calibration stability and interpretability, making it the most suitable model for deployment within an operational food price monitoring framework.	See the abstract section